

torch.linalg.norm#

<https://pytorch.org/docs/stable/generated/torch.linalg.norm.html>

Description:

Computes a vector or matrix norm. Supports input of float, double, cfloat and cdouble dtypes. Whether this function computes a vector or matrix norm is determined as follows: If dim is an int, the vector norm will be computed. If dim is a 2-tuple, the matrix norm will be computed.

Function Signature

```
torch.linalg.norm(A, ord=None, dim=None, keepdim=False, *,  
out=None, dtype=None) → Tensor#
```

Parameters

Keyword Arguments

Returns

Parameters

Keyword

out (Tensor, optional) – output tensor. Ignored if None. Default: None. dtype (torch.dtype, optional) – If specified, the input tensor is cast to dtype before performing the operation, and the returned tensor's type will be dtype. Default: None

Returns:

A real-valued tensor, even when A is complex.

Code Examples

```
>>> from torch import linalg as LA
>>> a = torch.arange(9, dtype=torch.float) - 4
>>> a
tensor([-4., -3., -2., -1.,  0.,  1.,  2.,  3.,  4.])
>>> B = a.reshape((3, 3))
>>> B
tensor([[ -4.,  -3.,  -2.],
        [ -1.,   0.,   1.],
        [  2.,   3.,   4.]])

>>> LA.norm(a)
tensor(7.7460)
>>> LA.norm(B)
```

```
tensor(7.7460)
>>> LA.norm(B, 'fro')
tensor(7.7460)
>>> LA.norm(a, float('inf'))
tensor(4.)
>>> LA.norm(B, float('inf'))
tensor(9.)
>>> LA.norm(a, -float('inf'))
tensor(0.)
>>> LA.norm(B, -float('inf'))
tensor(2.)

>>> LA.norm(a, 1)
tensor(20.)
>>> LA.norm(B, 1)
tensor(7.)
>>> LA.norm(a, -1)
tensor(0.)
>>> LA.norm(B, -1)
tensor(6.)
>>> LA.norm(a, 2)
tensor(7.7460)
>>> LA.norm(B, 2)
tensor(7.3485)

>>> LA.norm(a, -2)
tensor(0.)
>>> LA.norm(B.double(), -2)
tensor(1.8570e-16, dtype=torch.float64)
>>> LA.norm(a, 3)
tensor(5.8480)
>>> LA.norm(a, -3)
tensor(0.)
```

```
>>> c = torch.tensor([[1., 2., 3.],  
...                  [-1, 1, 4]])  
>>> LA.norm(c, dim=0)  
tensor([1.4142, 2.2361, 5.0000])  
>>> LA.norm(c, dim=1)  
tensor([3.7417, 4.2426])  
>>> LA.norm(c, ord=1, dim=1)  
tensor([6., 6.])
```

```
>>> A = torch.arange(8, dtype=torch.float).reshape(2, 2, 2)  
>>> LA.norm(A, dim=(1,2))  
tensor([ 3.7417, 11.2250])  
>>> LA.norm(A[0, :, :]), LA.norm(A[1, :, :])  
(tensor(3.7417), tensor(11.2250))
```

Notes & Warnings

NOTE See also `torch.linalg.vector_norm()` computes a vector norm. `torch.linalg.matrix_norm()` computes a matrix norm. The above functions are often clearer and more flexible than using `torch.linalg.norm()`. For example, `torch.linalg.norm(A, ord=1, dim=(0, 1))` always computes a matrix norm, but with `torch.linalg.vector_norm(A, ord=1, dim=(0, 1))` it is possible to compute a vector norm over the two dimensions.

Detailed Documentation

Documentation

Computes a vector or matrix norm.

Supports input of float, double, cfloat and cdouble dtypes.

Whether this function computes a vector or matrix norm is determined as follows:

If dim is an int, the vector norm will be computed.

If dim is a 2-tuple, the matrix norm will be computed.

If dim= None and ord= None, A will be flattened to 1D and the 2-norm of the resulting vector will be computed.

If dim= None and ord != None, A must be 1D or 2D.

ord defines the norm that is computed. The following norms are supported:

$\max(\sum(\text{abs}(x), \text{dim}=1))$

$\min(\sum(\text{abs}(x), \text{dim}=1))$

$\max(\sum(\text{abs}(x), \text{dim}=0))$

$\min(\sum(\text{abs}(x), \text{dim}=0))$

largest singular value

smallest singular value

$\sum(\text{abs}(x)^{\{\text{ord}\}})^{\{(1 / \text{ord})\}}$

where inf refers to float('inf'), NumPy's inf object, or any equivalent object.

`torch.linalg.vector_norm()` computes a vector norm.

`torch.linalg.matrix_norm()` computes a matrix norm.

The above functions are often clearer and more flexible than using `torch.linalg.norm()`. For example, `torch.linalg.norm(A, ord=1, dim=(0, 1))` always computes a matrix norm, but with `torch.linalg.vector_norm(A, ord=1, dim=(0, 1))` it is possible to compute a vector norm over the two dimensions.

`A (Tensor)` – tensor of shape `(*, n)` or `(*, m, n)` where `*` is zero or more batch dimensions

`ord (int, float, inf, -inf, 'fro', 'nuc', optional)` – order of norm. Default: None

`dim (int, Tuple[int], optional)` – dimensions over which to compute the vector or matrix norm. See above for the behavior when `dim= None`. Default: None

`keepdim (bool, optional)` – If set to `True`, the reduced dimensions are retained in the result as dimensions with size one. Default: `False`

`out (Tensor, optional)` – output tensor. Ignored if `None`. Default: `None`.

`dtype (torch.dtype, optional)` – If specified, the input tensor is cast to `dtype` before performing the operation, and the returned tensor's type will be `dtype`. Default: `None`

A real-valued tensor, even when `A` is complex.

Using the `dim` argument to compute vector norms:

Using the `dim` argument to compute matrix norms:

